



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2022-2023
PRE-BOARD EXAMINATION [SET-A]

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

FULL MARKS: 100
TIME: 2 HRS

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

This paper consist of ten pages.

This Paper is divided into two Sections.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets[.].

SECTION A(40 Marks)

(Attempt all questions from this Section.)

Question 1

[1x20=20]

Choose the correct answer from the following:

(i) Which of the following modifier can be applied to a variable declaration to ensure that the value stored in a variable cannot be changed after the variable has been initialized.

- (a) const (b) final (c) final const (d) static

(ii) What will be the value of x after executing the following statement, if int x=15, y=20

$x=(x<y)?(y+x) : (y-x)$

- (a) 55 (b) 36 (c) 28 (d) 35

(iii) What would happen if a constructor has a return type?

- (a) Compilation error (b) Runtime error
(c) Compiles and runs successfully (d) Illegal access error

(iv) Predict the output: `System.out.println(Math.ceil(3.4)+Math.pow(2,3));`

- (a) 13 (b) 11.0 (c) 13.0 (d) 12.0

(v) class Output

```
{ public static void main()
{ int a = 1; int b = 2; int c; int d;
  c = ++b;   d = a++;
  c++; b++; ++a;
  System.out.println(a + " " + b + " " + c); } }
```

(a) 3 2 4 (b) 3 2 3 (c) 2 3 4 (d) 3 4 4

(vi) What will be the output of the following Java code snippet if “AMIT” and “AMAN” are passed to the function.

```
void test(String x, String y)
{ if(x.compareTo(y)>0)
  System.out.println(x);
  else System.out.println(y); }
```

(a) AMIT (b) AMAN (c) compilation error (d) runtime error

(vii) Analyse the program snippet and mention the number of times the loop is executed.

```
int a,b;
for(a=6, b=4;a<=4;a=a+6)
{ if(a%b==0) break; }
System.out.println(a);
```

(a) 0 (b) 6 (c) 4 (d) 1

(viii) What is the process of defining more than one method in a class differentiated by method signature?

(a) Function overriding (b) Function overloading
(c) Function doubling (d) Function repeating

(ix) What does the following code print?

```
int a, b;
for(a=1;a<=3;a++)
{ for(b=(70+a); b<=71; b++)
  { System.out.print((char)b); }
  System.out.println( ); }
```

(a) g (b) F (c) G (d) h

(x) When we pass an array to a method, the method receives _____.

- (a) A copy of the array (b) A copy of the first element
(c) The reference of the array. (d) The length of the array.

(xi) Consider the following String array and give the output:

```
String a[]={“DELHI”, “CHENNAI”, “MUMBAI”, “LUCKNOW”, JAIPUR”};
```

```
System.out.print(a[0].length()>a[3].length());
```

```
System.out.print(a[4].substring(0,3));
```

- (a) trueJAIP (b) trueJAI (c) falseJAIP (d) falseJAI

(xii) Predict the output of the following code:-

```
public class Test
```

```
{ public static void main( )
```

```
{ int a[]=new int[5];
```

```
  a[0]=4; a[1]=8; a[2]=7; a[3]=12; a[4]=3;
```

```
  System.out.println(a[2+3]); } }
```

- (a) 712 (b) 19 (c) compilation error (d) runtime error

(xiii) Predict the output:-

```
class output
```

```
{ public static void main()
```

```
{ double a, b,c;
```

```
  a = 3%4;    b = 0/5;    c=0%6;
```

```
  System.out.println(a);
```

```
  System.out.println(b);
```

```
  System.out.println(c); } }
```

- | | | | |
|---------|----------|-------|-------|
| (a) 3.0 | (b) 3.0 | (c) 3 | (d) 3 |
| 0.0 | 0.0 | 0 | 0.0 |
| 0.0 | Infinity | NaN | 0.0 |

(xiv) Predict the output:

```
int Integer = 24; char String = ‘I’;
```

```
System.out.print(Integer);
```

```
System.out.print(String);
```

- (a) compile error (b) throws exception (c) I (d) 24I

(xv) Which of the following statement is the most appropriate for the private members?

- (a) They are visible out of the class in which they are defined.
- (b) They can be used in the sub-classes.
- (c) They are only visible in the class in which they are declared.
- (d) All of the above.

(xvi) What will be the output of the following Java code with many constructors?

```
public class Constructor7
{
    Constructor7(int a)
    {
        System.out.println("Book=" + a);
    }
    Constructor7(float a)
    {
        System.out.println("Pen="+ a);
    }
    public static void main()
    {
        Constructor7 con = new Constructor7(50.5f);
    }
}
```

- (a) Book=50 (b) Pen=50.5 (c) Compiler error (d) Pen=50.5f

(xvii) Predict the output:-

```
class char_increment
{
    public static void main( )
    {
        char c1= 'D';   char c2=84;
        c2++;          c1++;
        System.out.println(c1+ " " + c2);
    }
}
```

- (a) E U (b) U E (c) V E (d) U F

(xviii) Predict the output:-

```
int [ ] items = {2, 7, 3, 5, 8, 9};
int funny = items[0];
for (int i = 0; i<items.length; i++)
{
    if (items[i ] > funny)
        funny = items[i ];
}
System.out.print(funny);
```

- (a) 2 (b) 9 (c) 2 3 5 7 8 9 (d) 2 7 3 5 8 9

(xix) class box

```
{ int width;    int height;
  int length;   int volume;
  void volume(int height, int length, int width)
  {   volume = width*height*length;   }
  public static void main()
  {   box obj = new box();
      obj.volume(3,2,1);
      System.out.println(obj.volume); } }
```

(a) 0 (b) 1 (c) 6 (d) 25

(xx) Predict the output:

```
String [ ] name = { "Joe", "Sue", "Tom", "Jill", "Patty"};
for (int i = 0; i<name.length; i = i + 2)
System.out.print(name[i] + " ");
```

(a) Joe Sue Tom Jill Patty (b) Patty Jill Tom Sue Joe
(c) Joe Tom Patty (d) Sue Jill

Question 2

(i) Name the following: -

[2]

- a. Variables which are created once for the entire class and are shared by all the objects.
- b. The construct which accomplishes “fall through”.

(ii) Identify the errors and rewrite the correct code:-

[2]

```
class Sample
{   public static void main( )
    {   int n,p;    float k,r;    n=25; p=12;
        if(n=25)
        {   k=Math.pow(p,2);
            System.out.println(“The value of”+p+”=” +k );   }
        else
        {   r=Math.cbrt(n);
            System.out.println(“The value of “+n+ “=”+r); } } }
```

(iii) “Array is a composite data type.” Explain the statement.

[2]

(iv) Why do we need a constructor as a class member? How are constructors invoked?

[2]

(v) Give the output of the following program code when the below function will be called

as: (a) f(28,39) (b)f(27,39)

[2]

```
public boolean f(int a , int b)
{
    boolean e=false;
    while(a>1 && b>1)
    {
        if(a>b)
            a=a-b;
        else
            b=b-a;
    }
    if (a==1|| b ==1)
        e=true;
    return e; }

```

(vi) In the program given below, state the name and the value of the following:

a. class variable b.local variable

[2]

```
class Myclass
{
    static int x=7;
    int y=2;
    void sampleMethod(int n)
    {
        System.out.println(n);
        System.out.println(y); }
    public static void main( )
    {
        Myclass obj=new Myclass( );
        System.out.println(x);
        obj.sampleMethod(5);
        int a=6;
        System.out.println(a); }
}

```

(vii) Predict the output of the following program segment:

[2]

```
if int x[] = {4,3,7,8,9,10};
```

what are the values of p and q?

a. int p=x[1]% x[0]+ ++x[3]* x[3]/x[5]++;

b. int q=x[2]+x[5]/x[1]*x[2];

(viii) Answer the following questions:-

[2]

```
class sum  
  
{ private int x, y;  
  
    public int total;  
  
    sum(int a,int b)  
  
    {   x=a;  
  
        y=b;  
  
        total=x+y;}  
  
    public static void main( )  
  
    {   sum tot= new sum(50,46);  
  
        System.out.print(tot.total); } }
```

a.How can we access the variables x and y within the main method? Explain.

b.What type of constructor is sum (.....)?

(ix) Consider the following function and predict the output for the following:

[2]

```
public void show (String S)  
  
{ if(S.length( )>10)  
  
    System.out.println(S.substring(11,S.length( )));  
  
    else  
  
    System.out.println(S.toLowerCase());}
```

a.If the input to the function is: “ This is a Program”.

b. If the input to the function is “CAPTAIN”.

(x) Rewrite the program segment using if...else instead of the ternary operator.

[2]

```
String grade=(mark>=90)? “A” : (mark>=80) ? “B” : “C”;
```

Section B (60 Marks)

(Answer any four questions from this Section.)

The answers in this section should consist of the programs in either BlueJ environment or any program environment with java as the base.

Each program should be written using variable description/mnemonic code so that the logic of the program is clearly depicted. Flowcharts and algorithms are not required.

Question 3

[15]

Define a class *PhoneBill* that calculates the telephone bill of a consumer with the following description:

Data Members/Instance Variables:

String name : name of the consumer
long phoneNumber :phone number of the consumer
int billno :bill number
int call :number of calls consumed in a month
double amount :bill amount to be paid by the person

Member Functions:

- i.PhoneBill() : A constructor to initialize data members with their default values
- ii.PhoneBill(.....) : A parameterized constructor to accept the billno,name and no of calls consumed
- iii.void calculate() : This member function calculates the monthly telephone bill for a consumer as per the table given below:

No_of_calls	Rate
First 100 calls	0.60 per call
Next 100 calls	0.80 per call
Next 100 calls	1.20 per call
Above 300 calls	1.50 per call

- iv. void display() :to display the values of all the data members with the final billing amount where a fixed monthly rental applicable to all consumers is Rs 125.

Write the main method to invoke all the above functions to perform the desired task.

Question 4**[15]**

Write a program in Java to accept two Single Dimensional Array, Arr1[] and Arr2[]. Arr1[] stores 6 elements in it and Arr2[] to store 4 elements in it. Join the elements of both these arrays in a third array Arr3[] where Arr1[] elements are followed by Arr2[].

INPUT Arr1[]	INPUT Arr2[]	OUTPUT Arr3[]
4	19	4
6	23	6
1	7	1
2	8	2
3		3
10		10
		19
		23
		7
		8

Then search for an element from Arr3[] using Binary Search technique. If the element is found display the element along with its position, otherwise display the message “Search element is not found”.

Question 5**[15]**

Define a class to overload the method *check()* as follows:

i. void check(String str, char ch)- to find and print the frequency of a character in a string

Sample Input: str= “success” ch= ‘s’ Sample Output: Number of s present is =3

ii. void check(String s1) – to display only the vowels from string s1, after converting it to lowercase. Sample Input: s1= “computer” Sample Output: oue

iii. void check(String s, String s1) – checks and prints whether both the words are equal or not

Write the main method to invoke the overloaded functions.

Question 6 [15]

A Tech Number has even number of digits. If the number is broken into two equal halves, then the square of sum of these halves is equal to the number itself. Write a program to input a number, check and print whether it is a Tech number or not.

Example: Consider the number 3025

Square of sum of the halves of 3025 $= (30+25)^2 = (55)^2 = 3025$.

Question 7 [15]

Design a user-defined method to perform the following:-

i. void display (char, char) : A method to accept two different characters and display the sum and difference of their ASCII values.

Sample input: A

D

Sample output: The sum of ASCII values = 133

The difference of ASCII values = -3

ii. void removevowel(String) : A method to accept a string/word and display the new string after removing all the vowels present in it.

Sample input: COMPUTER APPLICATIONS Sample output: CMPTR PPLCTNS

iii. void count(String) : A method to input a word /string and count all the alphabets excluding the vowels present in the string/word .

Sample input: Happy New Year Sample output: no of alphabets excluding vowels = 8

Question 8 : [15]

Write a program to generate the following patterns using functions.

2	Z Y X W V
4 6	Z Y X W
8 10 12	Z Y X
14 16 18 20	Z Y
22 24 26 28 30	Z

